Combinatorial Game Theory

Key Concepts, Theorems, and Proofs

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P-Positions and N-Positions

Definition

- **P-Position** — a position from which the player to move *cannot* force a win (a losing position).
- **N-Position** — a position from which the player to move *can* force a win (a winning position).

Theorem (Recursive Characterization)

1. Every terminal position (no legal moves) is a P-position.
2. A position is an N-position if some move leads to a P-position.
3. A position is a P-position if every move leads to an N-position.

Recursive Characterization — Proof

Proof.

By strong induction on the number of moves remaining (finite, since the game terminates).

A terminal position has no moves, so by convention the player to move loses — rule 1.

Assume the classification is correct for every position reachable in fewer moves. If some option is a P-position, the mover moves there, handing the opponent a losing position — rule 2.

If every option is an N-position, every move hands the opponent a winning position, so the mover loses — rule 3.

Nim: The Nim-Sum

Definition (Nim-Sum)

The nim-sum of heap sizes $a_1, \ldots, a_n$ is their bitwise XOR,

$$a_1 \oplus a_2 \oplus \cdots \oplus a_n. \tag{1}$$

Theorem (Bouton's Theorem)

A Nim position is a P-position if and only if

$$a_1 \oplus a_2 \oplus \cdots \oplus a_n = 0. \tag{2}$$

Bouton's Theorem — Proof

Proof.

Part I. Every move from a zero nim-sum position produces a non-zero nim-sum position. If $a_1 \oplus \cdots \oplus a_n = 0$ and a move changes exactly one heap a_k, the XOR value must change, so the new nim-sum cannot remain zero.

Part II. Every non-zero nim-sum position has a move to a zero nim-sum position. Let $S = a_1 \oplus \cdots \oplus a_n \neq 0$, and let the most significant set bit of S lie in heap a_i. Setting $a_i' = a_i \oplus S$ gives $a_i' < a_i$ (legal), and the new nim-sum is

$$S \oplus a_i \oplus (a_i \oplus S) = 0. \tag{3}$$

Combining Parts I and II: the nim-sum is zero iff the position is a P-position.

The Mex Function and Grundy Numbers

Definition (Mex)

For a set of non-negative integers S,

$$\text{mex}(S) = \min\{\, n \geq 0 : n \notin S \,\}. \tag{4}$$

Definition (Grundy Number)

Every impartial position P has Grundy number

$$g(P) = \text{mex}\{\, g(Q) : Q \text{ reachable from } P \,\}. \tag{5}$$

The Sprague–Grundy Theorem

Theorem (Sprague–Grundy)

$$P \cong *g(P) \tag{6}$$

Every finite impartial game is equivalent to a single Nim heap of size $g(P)$.

Proof.

Terminal positions have $g = 0$, matching an empty Nim heap. By definition of mex, P's options realize exactly the Grundy values $\{0, 1, \ldots, g(P) - 1\}$ among reachable values missing $g(P)$ itself — precisely the options available from a Nim heap of size $g(P)$. Since P/N classification depends only on reachable Grundy values, $P \cong *g(P)$ by induction on game length.

Sums of Games — Proof

Theorem (Sum Theorem)

If independent impartial games P, Q have Grundy values a, b, then $P + Q$ has Grundy value $a \oplus b$.

Proof.

Let $s = a \oplus b$. Since $a = \text{mex}(\cdot)$ over P's options, every value $0, \ldots, a-1$ occurs among them (else a would not be minimal), and a itself never does.

No move reaches s. A move changes $P \to P'$ or $Q \to Q'$, with $G(P') \neq a$ or $G(Q') \neq b$; by induction $G(P' + Q) = G(P') \oplus b \neq a \oplus b = s$ (similarly for Q).

Every $m < s$ is reachable. Let d be the top bit of s; it is set in exactly one of a, b, say a. Set $c = m \oplus b$; a bit-by-bit check at d shows $c < a$, so some option P' of P has $G(P') = c$, and $P' + Q$ has value $c \oplus b = m$. Hence mex of the reachable values is exactly s.

Sums of n Games

Theorem (Sums of Impartial Games)

$g = g_1 \oplus g_2 \oplus \cdots \oplus g_n$ for n independent components.

Proof.

By induction on n, using the Sum Theorem repeatedly. The case $n = 2$ is the Sum Theorem itself. For $n > 2$, group the first $n - 1$ components: by the inductive hypothesis they combine to Grundy value $g_1 \oplus \cdots \oplus g_{n-1}$; applying the Sum Theorem to this group and the n-th component gives

$$(g_1 \oplus \cdots \oplus g_{n-1}) \oplus g_n = g_1 \oplus \cdots \oplus g_n. \tag{7}$$

Nimbers

Definition

A Nim heap of n stones is denoted $*n$, a nimber. Every finite impartial position is equivalent to one nimber.

Theorem (Nimber Addition)

$$*a + *b = *(a \oplus b) \tag{8}$$

Proof.

$*a$ and $*b$ placed side by side form the disjunctive sum of two impartial games with Grundy values a and b. By the Sum Theorem, the combined position has Grundy value $a \oplus b$; by Sprague–Grundy, a position with Grundy value $a \oplus b$ equals $*(a \oplus b)$.

Wythoff Nim and the Golden Ratio

Theorem (Golden Ratio Characterization)

The n-th P-position pair (a_n, b_n) of Wythoff Nim is

$$a_n = \lfloor n\varphi \rfloor, \qquad b_n = \lfloor n\varphi^2 \rfloor, \qquad \varphi = \frac{1+\sqrt{5}}{2}. \tag{9}$$

Lemma (Beatty's Theorem)

If $\alpha, \beta > 1$ are irrational with $\frac{1}{\alpha} + \frac{1}{\beta} = 1$, the sequences $\lfloor n\alpha \rfloor$ and $\lfloor n\beta \rfloor$ $(n \geq 1)$ partition the positive integers.

Note φ satisfies $\varphi^2 = \varphi + 1$, so $\frac{1}{\varphi} + \frac{1}{\varphi^2} = \frac{\varphi+1}{\varphi^2} = \frac{\varphi^2}{\varphi^2} = 1$: Beatty's theorem applies to $\alpha = \varphi$, $\beta = \varphi^2$.

Wythoff Nim — Proof

Proof.

Since $b_n - a_n = \lfloor n\varphi^2 \rfloor - \lfloor n\varphi \rfloor = n$ (using $\varphi^2 = \varphi + 1$), each pair has the required gap structure. By Beatty's Theorem, $\{a_n\}$ and $\{b_n\}$ partition $\mathbb{Z}^+$, so every positive integer is some a_k or b_k for a unique k — hence every candidate P-position (x, y) with $x < y$ appears exactly once in the list, and every other position has x or y missing from its own list-pair, i.e. is not of this form.

No move connects two listed pairs. A move changes one heap (or both equally). Since $\{a_n\}, \{b_n\}$ partition $\mathbb{Z}^+$ with $a_n < b_n$ strictly increasing, decreasing either coordinate of (a_n, b_n), or both by the same amount, cannot land on another (a_k, b_k): this is verified directly from the strict monotonicity and irrationality of φ.

Every other position moves to a listed pair. A direct case check (reduce the larger heap, or the smaller, or both) shows any position off the list can reach some (a_n, b_n).

Hence $\{(a_n, b_n)\}$ satisfies both recursive P-position rules.

Green Hackenbush and Nim

Theorem

A bamboo stalk of length n is isomorphic to a Nim heap of size n.

Proof.

Removing the k-th edge from the bottom leaves a stalk of length $k-1$, so the options from a stalk of length n are $\{0, 1, \ldots, n-1\}$. Hence

$$g(n) = \text{mex}\{0, 1, \ldots, n-1\} = n. \tag{10}$$

Games, Outcome Classes, and Star

Definition (Conway Game)

$$G = \{ L \mid R \} \tag{11}$$

recursively, from the empty game $0 = \{ \mid \}$.

Outcome Classes

- $G > 0$ (Left wins) $G < 0$ (Right wins)
- $G = 0$ (2nd player wins) $G \parallel 0$ (1st player wins)

Example (Star)

$* = \{0 \mid 0\}$. Whoever moves first wins, so $* \parallel 0$.

Canonical Form and Disjunctive Sum

Definition (Canonical Form)

A game has no dominated options and no reversible moves — the simplest representative of its value.

Definition (Disjunctive Sum)

$G + H$: a move is a legal move in G or in H, leaving the other unchanged.

Numerical Values

Definition (Numerical Game Value)

$G = \{L \mid R\}$ has value $x \in \mathbb{R}$ if $L < x < R$ for all Left/Right options.

Theorem (Simplicity Rule)

The value of $G = \{L \mid R\}$ is the simplest number strictly between the largest Left option and the smallest Right option.

Simplicity Rule — Proof

Proof.

Let x be the simplest number with $L < x < R$. To show $G = x$, i.e. $G - x = 0$, we show the second player wins $G - x$.

If Left moves first in G, reaching some $L_i \leq L < x$, Right replies in $-x$ (or, if Left instead moved in $-x$ to a value $\geq x$, Right replies in G); this maintains a non-positive difference. Symmetrically if Right moves first.

No simpler x' works: if $L < x' < x$ (or $x < x' < R$) also satisfies $L < x' < R$, then x' — simpler and already sandwiched — would have been chosen instead, contradicting minimality of x.

Infinitesimals and the Up Game

Definition (Infinitesimal)

$-\varepsilon < G < \varepsilon$ for every positive real ε.

Theorem (Up Game)

$\uparrow = \{0 \mid *\}$, and $0 < \uparrow < \varepsilon$ for every positive real ε.

Proof.

Left moving first wins immediately (moves to 0); Right moving first can only reach $*$, where the first player (now Left) wins. So $\uparrow > 0$ always. Yet $\uparrow \neq 1$: in $1 = \{0 \mid\}$ Right has no move at all, whereas in $\uparrow$ Right's move to $*$ is real, making Left's edge strictly smaller than every positive real.

Tiny and Miny Games

Definition

For $x > 0$:

$$\text{Tiny}(x) = \{0 \mid -x\}, \qquad \text{Miny}(x) = \{-x \mid 0\} = -\,\text{Tiny}(x). \quad (12)$$

Tiny games are positive infinitesimals favoring Left; Miny games favor Right.

Mean Value, Temperature, Cooling

Definition (Mean Value)

For $G = \{L \mid R\}$: $\quad m = \dfrac{L + R}{2}$

Definition (Temperature)

For $G = \{L \mid R\}$: $\quad t = \dfrac{L - R}{2}$

Definition (Cooling)

$G_c = \{\, L - c \mid R + c \,\}$ — reduces temperature, preserves mean value; heating is the inverse.

Blue–Red–Blue Hackenbush Stack

Example

For a stack of Blue, Red, Blue edges (bottom to top) on the ground,

$$G = \left\{ 0, \tfrac{1}{2} \mid 1 \right\} = \frac{3}{4} \tag{13}$$

by the Simplicity Rule: the simplest number strictly between the largest Left option (1/2) and the smallest Right option (1).